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RESEARCH PAPERS

# Identification and Expression Profiling of a New β-amyrin Synthase Gene (*GmBAS3*) from Soybean[1]

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**Abstract**—Cyclization of 2,3-oxidosqualene by different oxidosqualene cyclase (OSC) genes is responsible for sapogenin heterogeneity. The very first phase is the conversion of 2,3-oxidosqualene into β-amyrin by β-amyrin synthase (BAS) gene, a member of OSC family, in soy saponin biosynthesis pathway. This paper reports the identification of a new BAS gene (*GmBAS3*) and its expression pattern in soybean (*Glycine max* (L.) Merr.). *GmBAS3* gene was identified by PCR/RACE method with an open reading frame of 2286 bp nucleotides encoding a 762 amino acid long protein devouring a characteristic QW motif repeated five times and DCTAE motif. *GmBAS3* shared 96 and 92% homology with *Glycyrrhiza uralensis* BAS and *Lotus japonicus* putative BAS respectively. Expression of the gene was detected by RT-PCR in regard to seedlings age and tissue type. A spatio-temporal expression of *GmBAS3* was found in 21-day-old seedlings in the hypocotyls, young leaves and mature leaves but not observed in stem and root tissues. No expression was perceived in 10-day-old seedling. This study also support the premise that β-amyrin synthesis hang on more than one type of BAS genes with there expression in different plant parts at different times.

*Keywords: Glycine max,* 2,3-oxidosqualene, cyclization, OSC genes, β-amyrin synthase, soyasaponin biosynthesis

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## INTRODUCTION

Plants produce structurally diverse array of saponins comprising of steroid or triterpene aglycone attached with one or more oligosaccharide sugar moieties [1]. The first step is the cyclization of 2,3-oxidosqualene in the biosynthesis of triterpenoid saponins and phytosterols pathway [2]. Enzymes carrying these cyclizations are labelled as oxidosqualene cyclases (OSCs, EC 5.4.99.x). Oxidosqualene cyclase catalysis mostly ensues according to the biogenetic isoprene regulation [3, 4]. The distribution and abundance of sapogenin types in plants varies and most sapogenins are neither limited to different plant orders nor do their dispersals noticeably follow the phylogeny of plants [5]. Definitely the most lavish type of triterpenoid sapogenins in nature appears to be the oleananes resulting from the 2,3-oxidosqualene cyclization is β-amyrin. Fascinatingly, the capability to yield precursors of triterpenoidal sapogenins was displayed to be more pervasive amongst plants than the described distribution of the same saponins. For instance, numerous plants in the Brassicaceae family such as *Arabidopsis thaliana* and *Brassica oleracea* cultivars have been stated to store pentacyclic triterpenoids such as β-amyrin, α-amyrin, and lupeol [6–10]. Furthermore, the partners of the Brassicaceae family, other plants of diverse phylogenetic origin such as grape (*Vitis vinifera*) [11], castor bean (*Ricinus communis*) [12], olive (*Olea europaea*) [13], tomato (*Solanum lycopersicum*) [14], and soybean (*Glycine max*) [15] are well-known to collect oleanane, ursane, and lupane type aglycones.

The sum of well-known and biochemically categorized OSCs has ominously increased throughout the last decade [16]. Most of OSCs have been isolated by the use of degenerative primers based on significantly conserved motives and successive re-formation of the full-length ORF by rapid amplification of cDNA ends (RACE-PCR) in *Gentiana straminea* [17], *Artemisia annua* [18], *Bruguiera gymnohriza* [19] whereas others were identified based on selection of cDNA libraries in *Nigella sativa* [20] and *Avena stirgosa* [21]. However,

[1] The article is published in the original.

*Abbreviations*: BAS—β-amyrin synthase; BLAST—Basic Local Alignment Search Tool; MEGA—Molecular Evolutionary Genetics Analysis; OSC—oxidosqualene cyclase; YPD—yeast extract-peptone-dextrose.

analysis of the *A. thaliana* genome exposed 13 OSC genes in total, suggesting the presence of a number of more OSCs in plants that have been found by the sequence similarity-based approaches applied so far for *A. thaliana* [22, 23]. The identification and expression profiling of two putative β-amyrin synthase genes *GmBAS1* and *GmBAS2* in soybean expressed divergent arrays with respect to developmental phase and tissue specificity [24]. Realizing the importance of β-amyrin synthase and rationalizing the higher levels of saponins in comparison to other plants, we hypothesized that higher rate of saponin biosynthesis in soybean may not be necessarily dependent on a single β-amyrin synthase gene.

Here, we report cloning, expression and functional characterization of a new β-amyrin synthase gene (*GmBAS3*) from soybean. The new gene was confirmed by (i) expressing *GmBAS3* in lanosterol synthase-deficient yeast mutant, (ii) characterizing both QW and DCTAE motives of GmBAS3, and (iii) correlating the phylogenetic relationship of GmBAS3 with other plant known OSCs including soybean OSCs and its spatio-temporal expression.

## MATERIALS AND METHODS

**Sample preparation and RNA extraction.** Seeds of soybean (*Glycine max* (L.) Merr. cv. Taekwang) were germinated aseptically on Gamborg's B5 salt basal medium with vitamines [25] at 25 ± 1°C with cycles of 16 h light/8 h dark for 21 days. Seedlings of 10 and 21 days were dissected into hypocotyl, stem, root, young leaves, and mature leaves and were immediately immersed in liquid nitrogen and stored at –80°C until RNA extraction. The shoot tip of soybean seedling was used as explant for callus production on initiation medium containing Gamborg's B5 supplemented with 0.05 mg/L kinetin, 0.5 mg/L 2,4-dichlorophenoxyacetic acid (2,4-D), 30 g/L sucrose and 0.7% tissue culture grade agar. The pH was adjusted to 5.8 using 1 N KOH and maintained at 25 ± 1°C for 40 days on continuous light. After successful induction, callus was immersed in liquid nitrogen and stored at –80°C until RNA extraction. As described in manufacturer's manual, total RNA was extracted from 100 mg of each frozen tissue by RNAiso Plus solution (Takara, Japan) and purified using Fruit-mate purification reagent kit (Takara).

**Cloning of oxidosqualene cyclase.** Two primer pairs were employed for putative OSC sequence cloning: (i) $FP_7$: 5'-GATGGAGGATGGGGACTACA-3' and $RP_7$: 5'-CGAGCCATACCATGAACCAT-3', (ii) $FP_8$: 5'-TCAGAACAAGGATGGAGGATG-3' and $RP_8$: 5'-CCACGAGCCATACCATGAAC-3'. Primers were ordered and synthesized commercially by Bioneer, Inc. (Korea). The cDNA first strand was synthesized using Revert Aid Premium First Strand cDNA Synthesis kit (Thermo Scientific, Korea), according to the manufacturer's instructions. First strand cDNA was amplified using hot start Ex-*Taq* DNA polymerase (Takara) under the following conditions: 94°C for 1 min followed by 32 cycles each of 94°C for 30 s, 61°C for 30 s, 72°C for 90 s, with a final extension at 72°C for 10 min. The amplified product (900 bp) was purified using QIAquick gel purification kit (Qiagen, United States) and was ligated into pGEM-T easy vector (Promega, Korea). *Escherichia coli* DH5α competent cells were transformed with recombinant vectors by heat shock method [25]. The cloned plasmid DNA was prepared using QIAprep spin miniprep kit (Qiagen) and sequenced by the BigDye$^{TM}$ terminator method using an ABI 3730XL DNA analyzer at Bioneer. Five colonies were sequenced using T7 forward primer and Sp6 reverse primer for confirmation of the correct sequence.

**Rapid amplification of cDNA ends (RACE) PCR.** Both 3'- and 5'-RACE PCR were performed using "Invitrogen" 3'- and 5'-RACE system, respectively. In 3'-RACE PCR, 1 μg of total RNA was reverse transcribed using adaptor primer (AP: 5'-GGCCACGCGTCGACTAGTACTTTTTTTTTTTTTTTTT-3') followed by amplified using sense primer (PM-$F_1$: 5'-AGGTTTGTGGGTCCAATCAC-3') along with abridged universal amplification primer (AUAP: 5'-GGCCACGCGTCGACTAGTAC-3') under the following conditions: 1 min at 94°C followed by 30 cycles each of 45 s at 94°C, 45 s at 59.5°C, 90 s at 72°C, and final extension for 10 min at 72°C. In 5'-RACE PCR, 1 μg of total RNA was reverse transcribed using antisense primer (PM-$R_1$: 5'-CCATGCTGCTAAACCACCTT-3'). According to manufacturer's instructions, two nested PCR were carried out with AUAP primer and antisense primer PM-$R_2$ (5'-TCTCCGTTTGGATCTTCCAC-3') with the following conditions: 2 min at 94°C followed by 30 cycles each of 45 s at 94°C, 45 s at 61.5°C, 90 s at 72°C, and final extension for 10 min at 72°C. Amplified products were gel purified, cloned into pGEM-T easy vector (designated as pGEM-T-3 contig and pGEM-T-5 contig, respectively) and sequenced as described previously.

**Heterologous expression of oxidosqualene cyclase in yeast.** Full-length cDNA was cloned into the yeast expression vector pYES2 (Invitrogen) with a galactose inducible GAL1 promoter. The construct was propagated in *E. coli* DH5α cells. The pYES2 and purified construct were separately electroporated into electrocompetent cells of *Saccharomyces cerevisiae* GIL77 (a lanosterol synthase deficient strain). Electroporated cells were dispersed in 1 mL of Yeast extract-Peptone-Dextrose (YPD) medium and incubated at 30°C for 1 h. Transformed cells were selected on synthetic minimal defined medium lacking uracil (SC-U). Selected cells were induced for 24 h with galactose, potassium phosphate and hemin. Cells were precipitated by centrifugation at 5000 rpm and refluxed in the mixture of 20% KOH (v/v) in 50% methanol (v/v). After addition of a half volume of water, the total content (homogenate) was extracted twice with hexane. Extracts were evaporated using water bath at 40°C until white crude residue formed which was dissolved in 200 μL of chloroform. Chloroform extract was separated on silica gel ($SiO_2$) coated thin layer chromatography (TLC) plates (10 × 20 cm, Merck, Germany) with the lower phase of chloroform : methanol : water (65 : 35 : 10, v/v/v). Comparing with authenticated standard (β-amyrin), suspected cyclization spot was scratched from TLC plate, re-dissolved in 100 μL of chloroform and subjected to high performance liquid chromatography (HPLC) analysis. HPLC was performed using a Shimadzu LC-10 AVP system (Japan) equipped with a SUPER-ODS column (4.6 × 200 mm) and a UV-VIS detector. Authenticated standard containing 5% of water was used as mobile phase with a flow rate of 1.0 mL/min for 45 min at 40°C. Peaks were detected at 202 nm UV.

**Expression of *GmBAS3* in soybean tissues.** Reverse transcription PCR (RT-PCR) was performed as mentioned earlier to examine the expression of *GmBAS3* in 40-day-old callus and in hypocotyl, stem, root, young leaves, and mature leaves of 10-day- and 21-day-old soybean seedlings. The RT-PCR experiments were conducted using the primers PM-$F_3$ (5'-TCACCAATGTGCAAAGGAAG-3') and PM-$R_2$ (5'-TCTCCGTTTGGATCTTCCAC-3') with the denaturation temperature 94°C for 2 min, 30 cycles each at 94°C for 45 s, 60.5°C for 45 s, and 72°C for 90 s, and finally 72°C for 10 min.

**Phylogenetic analysis.** Phylogenetic relationship of cloned *GmBAS3* with 27 known characterized plant OSCs and with 8 known uncharacterized soybean OSCs was inferred, based on the neighbor-joining method using the amino acid sequence. A phylogenetic tree was created using Molecular Evolutionary Genetics Analysis (MEGA) v. 5 software [26].

## RESULTS AND DISCUSSION

The phylogeny of OSCs advocates autonomous development of cyclases with analogous functionalities in different plant species, and numerous OSCs are identified to give rise to more than one cyclization product [27]. Reciprocally, findings designate the potential of OSCs to shift their product array somewhat speedily during evolution and thus provide plants with sporadic types of oxidosqualene cyclization yields. The existence of triterpenoids such as β-amyrin, α-amyrin, and lupeol in plants that do not produce analogous saponins additionally supports this hypothesis. The planned convergent evolution of metabolic pathways expanded to amassing of triterpenoid saponins, and the putatively swerved phylogenetic origin of the enzymes tangled, obstructs the forecast of yet undiscovered genes by sequence resemblance. Few reports on plant OSCs appreciates homology-based gene discovery strategies; such as the use of degenerative primers [23, 24, 28, 29], hybridization-based cDNA library screenings [26, 30] and excavation of genomic or transcriptomic datasets. Nevertheless, mining the OSC functionality needs heterologous expression and biochemical characterization of the enzymes. Although the precise biosynthetic pathway of soyasaponins is not elucidated yet, it has been shown that cyclization of 2,3-oxidosqualene into β-amyrin by β-amyrin synthase via dammarenyl cation is the first steadfast step in soyasaponin biosynthesis (Fig. 1). Soybean genome data mining revealed the presence of several potential OSCs transcripts [4]; however, the detailed experimental evidences for such transcripts are absent. We employed homology-based gene discovery stratagem by using degenerative primers [17–19, 26, 28, 30, 31] and excavation of genomic datasets [27, 29]. Based on the reproducibility two primer pairs (FP7 and RP7; FP8 and RP8) were employed to amplify 900 bp cDNA prepared from the mRNA of hypocotyl of 21-day-old seedling. Amplified sequences were purified, ligated into pGEM-T Easy vector and cloned in *E. coli* DH5α cells. The 3'- and 5'-RACE PCR was used to extend the core sequences. Resulting nucleotide segments (5'-terminal sequence, 3'-terminal sequence and a core sequence) were cloned, sequenced and organized into three contigs. Full length cDNAs were obtained by using DNA Baser v. 3.5.3 software and were deduced into amino acid sequence by ExPASY. Multiple sequence alignment by MEGA and BLAST (Basic local alignment search tool) search revealed that the produced cDNAs shared 96% similarity with *G. uralensis* β-amyrin synthase gene, 92% similarity with *L. japonicus* putative β-amyrin synthase and 90% similarity with *Medicago truncatula* β-amyrin synthase. The nucleotide sequence was submitted in GenBank (Accession ID: KF597523). Existence of QW motif and DCTAE motif assured that the sequences belongs to the OSC family. We assigned the new gene as *GmBAS3* (*G. max* β-amyrin synthase 3). The *GmBAS3* open reading frame consists of 2286 bp nucleotides encoding 762 amino acid long protein of around 87.33 kD. The amino acid sequence of GmBAS3 possessed a characteristic QW motif repeated five times and a DCTAE motif. Phylogenetic analysis among the reported OSCs of soybean showed that *GmBAS3* fall into a clade consisting entirely of β-amyrin synthases. Among soybean OSCs, *GmBAS3* was closely related to *GmBAS1* and farly related to cycloartenol synthase like isoform 1 (Fig. 2). Characteristically, all β-amyrin synthases from soybean (including the predicted genes) contained 5 QW motifs [4]; however, the QW motif positions and the 5 amino acids between Q and W residues were varied (table).

Expression of *GmBAS3* in lanosterol synthase deficient yeast resulted in β-amyrin production and was authenticated by TLC with standard β-amyrin but was not present in extracts of cells that had been trans-

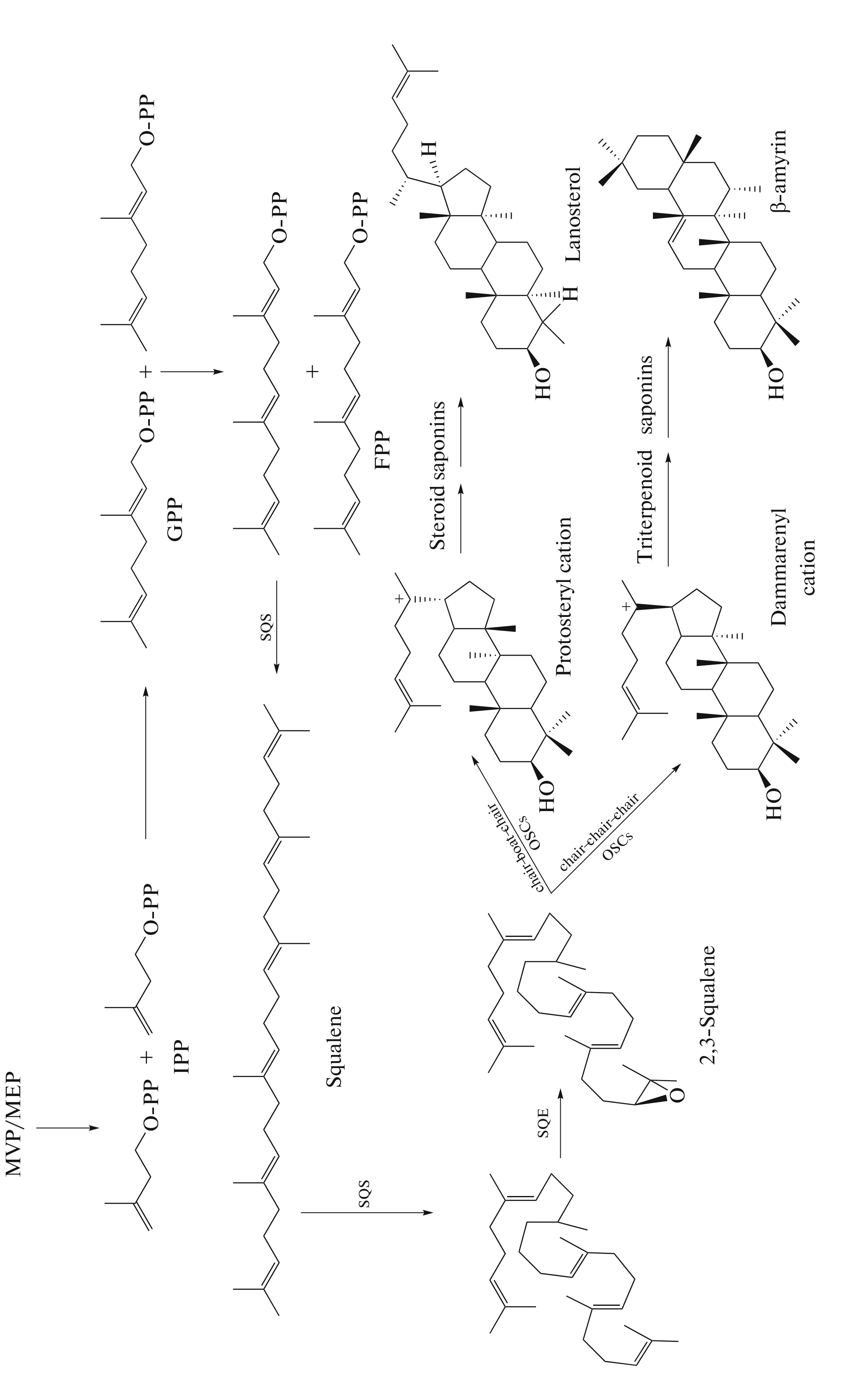


**Fig. 1.** Early steps in biosynthesis of phytosterols and saponins. IPP—isopentenyl pyrophosphate; GPP—geranyl pyrophosphate; FPP—farnesyl pyrophosphate; SQS—squalene synthase; SQC—squalene cyclase; SQE—squalene epoxidase; OSC—oxidosqualene cyclase.

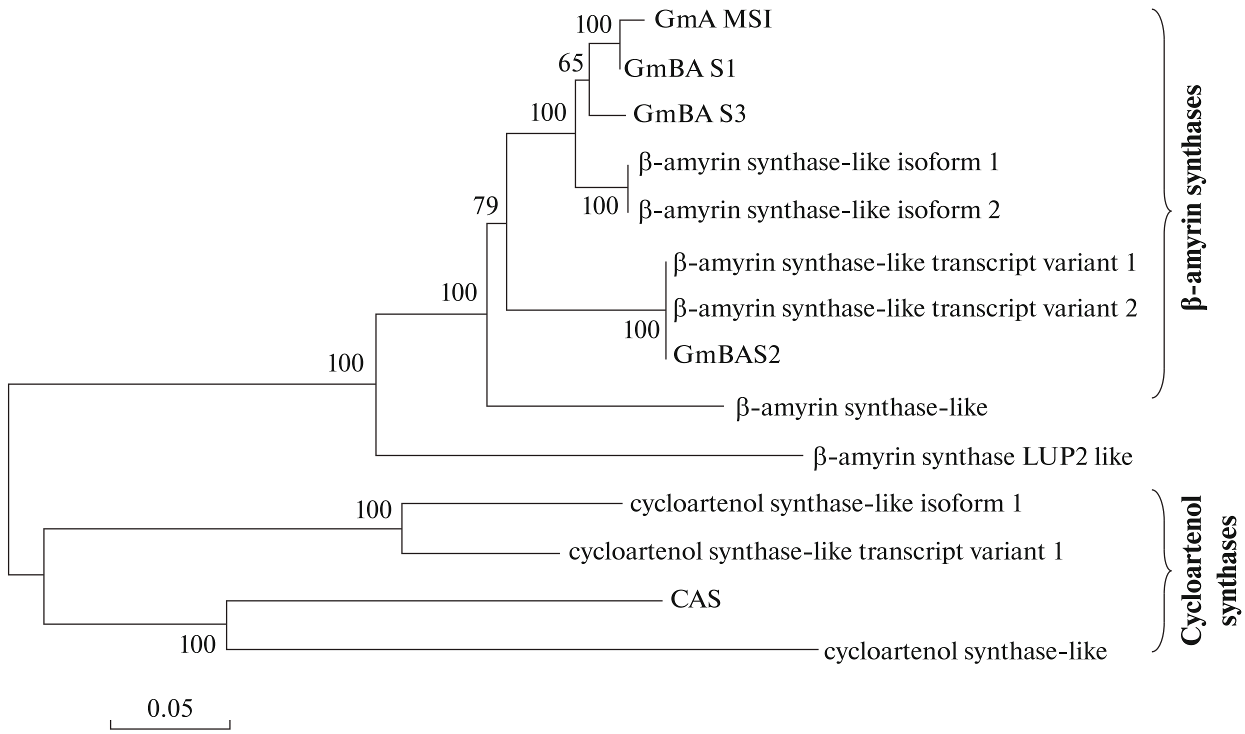


**Fig. 2.** Phylogenetic relationship of soybean OSCs by the neighbour-joining (NJ) method using the MEGA v. 5 program based on their amino acid sequences. Bootstrapping with 1000 replications was carried out for the NJ analysis. Numbers above branches are bootstrap values (%).

formed with the empty vector. HPLC analysis indicated retention time of novel product was identical to β-amyrin (Fig. 3). Expression of *GmBAS3* was spatio-temporal during soybean seedling growth stage and tissue of type. Fluctuating levels of *GmBAS* have been observed in soybeans [24]. Expression in seedlings was

Comparison of QW and DCTAE motifs among soybean BASs

| Soybean BASs | QW motif's amino acid sequence and its location in respective genes | | | | | DCTAE motif's location |
|---|---|---|---|---|---|---|
| | 1st | 2nd | 3rd | 4th | 5th | |
| GmAMS1 | QTSDGHW | QNEDGGW | QTADGSW | QREDGGW | QLEEGDW | 462–466 |
| | 109–115 | 159–165 | 579–585 | 628–634 | 693–699 | |
| GmBAS1 | QTSDGHW | QNEDGGW | QTADGSW | QREDGGW | QLEEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |
| GmBAS2 | QTSDGHW | QNEDGGW | QTTNGSW | QREDGGW | QLEEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |
| GmBAS3 | QTSDGHW | QNMDGGW | QTADGSW | QREDGGW | QLEEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |
| XP_003531781.1 | QTSDGHW | QNEDGGW | QTADGSW | QREDGGW | QLEEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |
| XP_003521100.1 | QTSDGHW | QNEDGGW | QTTNGSW | QREDGGW | QLEEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |
| XP_003521101.1 | QTSDGHW | QNEDGGW | QTTNGSW | QREDGGW | QLEEGDW | 462–466 |
| | 109–115 | 159–165 | 579–585 | 628–634 | 693–699 | |
| XP_003531782.1 | QTSDGHW | QNEDGGW | QTADGSW | QREDGGW | QLEEGDW | 462–466 |
| | 109–115 | 159–165 | 579–585 | 628–634 | 693–699 | |
| XP_003521102.1 | QTSDGHW | QNEDGGW | QTANGSW | QKEDGGW | QLKEGDW | 485–489 |
| | 109–115 | 159–165 | 602–608 | 651–657 | 713–719 | |

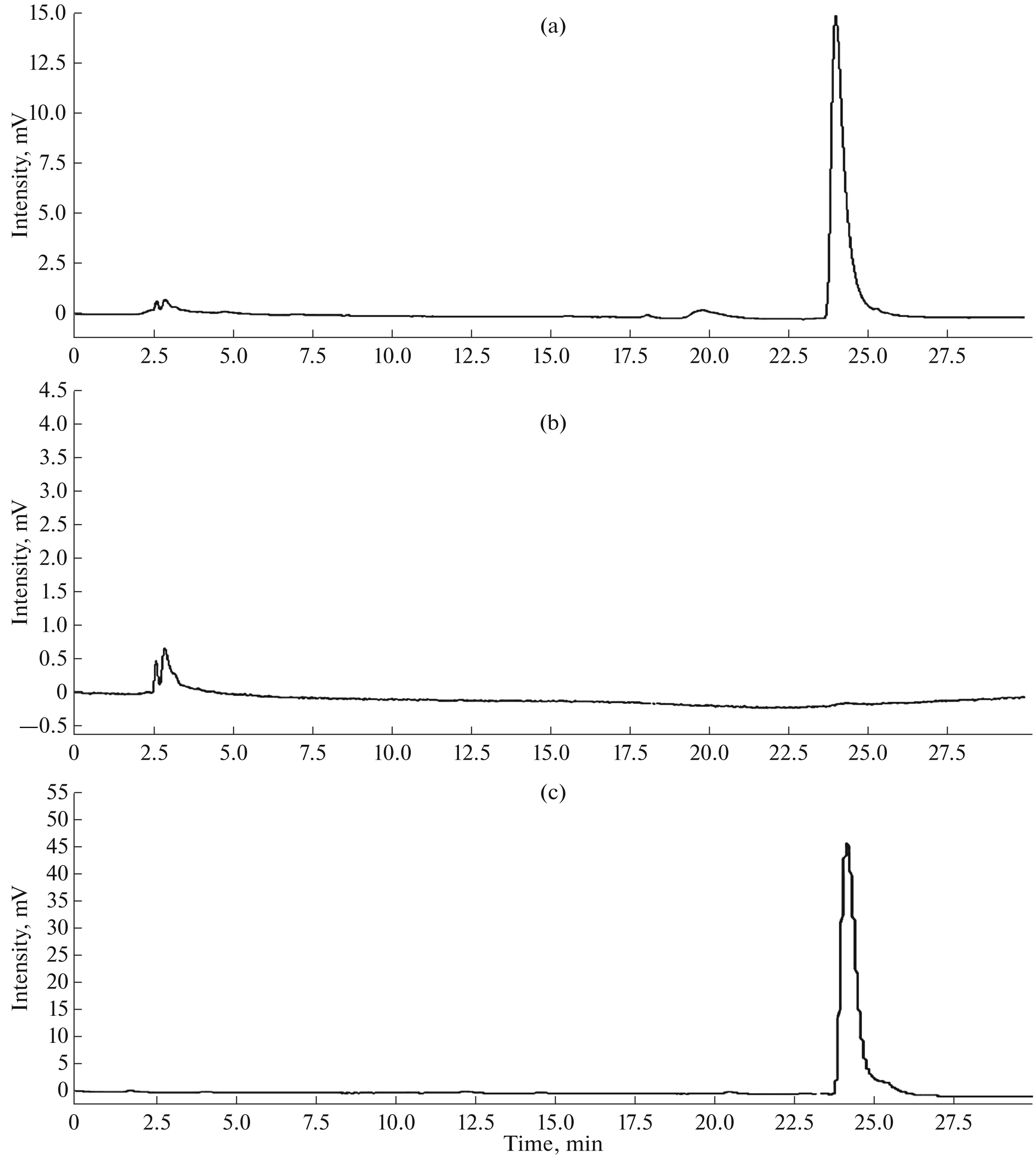


**Fig. 3.** HPLC chromatogram of purified product extracted from GIL77 mutant yeast cells transformed with pYES-BAS construct. a—β-amyrin standard; b—extracts from yeast cells having empty vector; c—extracts from yeast cells having pYES-BAS construct. Peaks were detected at 202 nm UV.

found to be tissue-specific and age-dependent (Fig. 4). Although the *GmBAS3* transcripts were not detected in any tissues of 10-day-old seedlings, however, 21-day-old seedlings showed tissue specificity in relation to *GmBAS3* expression. Seedlings of 21-day age showed noticeable *GmBAS3* expression in hypocotyls, young leaves, and mature leaves but there was no detectable expression in stems and roots.

Race to the discovery of related genes exhibiting evolutionary conserved sequences should continue to completely explain the biochemical pathway of sapogenin synthesis by employing recently emerged high throughput next generation sequencing techniques amalgamated with protein refinement assays and mass spectrometry. Furthermore, utilization of pile of available datasets can aid the putative gene function discov-

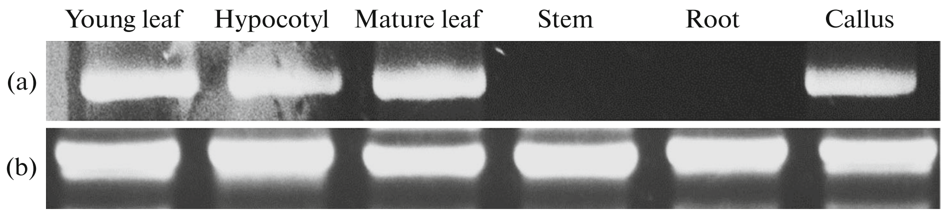


**Fig. 4.** Qualitative expression of *GmBAS3* in different tissues of 21-day-old soybean seedlings by reverse transcription PCR (RT-PCR). a—expression pattern of *GmBAS3*; b—amplified 18S RNA.

ery by employing recently developed bioinformatics tools. Development of genetic maps and related databases have to accelerate gene discovery.


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